

ATBASAR, Kazakhstan

WMO: 350780

Lat: 51.8419N

Lon: 68.3636E

Elev: 997

StdP: 14.17

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-23.4	-33.7	0.8	-28.3	-28.5	1.2	-23.1	36.9	13.6	32.4	15.7	6.5	200	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	90.1	63.6	86.6	62.9	83.3	62.0	67.3	81.3	65.9	80.3	64.5	78.2	11.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	89.3	71.2	61.3	84.1	70.2	59.5	78.8	69.1	32.3	81.1	31.2	80.5	30.1	77.8	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.4	26.8	23.0	DB	-34.7	95.9	5.6	3.7	-38.7	98.5	-42.0	100.7	-45.2	102.8	-49.3	105.6	(4)
(5)			WB	-34.0	70.0	5.9	2.2	-38.2	71.6	-41.7	72.9	-45.0	74.1	-49.3	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	0.5	3.1	17.3	41.5	55.5	64.8	67.2	64.9	53.5	39.7	20.6	7.0		
	DBStd	26.68	14.09	12.54	12.43	11.42	8.65	7.51	6.43	7.78	8.97	9.28	13.96	13.89		
	HDD50	6824	1534	1312	1013	297	45	3	0	2	62	339	883	1334		
	HDD65	10744	1999	1732	1478	706	309	93	51	100	359	785	1333	1799		
	CDD50	1887	0	0	0	41	216	447	533	463	168	19	0	0		
	CDD65	330	0	0	0	0	14	87	119	96	14	0	0	0		
	CDH74	4381	0	0	0	18	311	1174	1388	1246	239	5	0	0		
	CDH80	1506	0	0	0	1	68	407	470	479	81	0	0	0		
	WSAvg	10.6	10.6	11.7	11.9	11.2	11.2	10.1	9.3	9.0	9.5	10.1	10.9	11.4		
	PrecAvg	12.0	0.9	0.6	0.6	0.7	1.2	1.5	1.9	1.3	0.9	1.0	0.8	0.9		
	PrecMax	17.9	3.3	2.4	1.8	1.9	2.8	6.1	6.0	3.6	2.3	3.4	2.2	2.3		
(15) Precipitation	PrecMin	6.1	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.1		
	PrecStd	2.3	0.6	0.4	0.4	0.5	0.6	1.1	1.3	0.9	0.6	0.7	0.6	0.5		
	DB	32.3	32.2	46.8	76.4	86.7	92.9	94.0	94.5	89.1	72.3	48.8	33.1			
	MCWB	31.7	31.4	41.0	55.1	58.9	64.2	66.0	64.1	60.4	51.1	43.0	31.8			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	29.2	29.6	38.3	70.3	81.3	88.3	88.9	89.7	81.1	65.4	43.3	31.5			
	2% MCWB	28.4	28.8	35.8	52.1	58.0	63.4	64.4	62.9	57.2	49.7	40.1	30.8			
	5% DB	25.4	26.3	35.7	65.5	77.3	84.9	85.2	85.5	74.7	59.9	39.3	28.7			
	5% MCWB	24.6	25.3	33.9	50.6	56.6	62.7	63.7	62.7	55.3	46.9	36.8	27.9			
	10% DB	22.1	22.9	33.5	60.0	73.6	81.0	81.6	81.3	69.7	54.4	36.0	25.7			
	10% MCWB	21.2	22.1	32.3	48.0	55.2	61.3	63.2	61.5	53.5	44.7	34.2	24.8			
	0.4% WB	31.8	31.5	41.7	57.6	63.2	68.4	69.2	67.6	62.7	53.8	44.6	32.3			
	0.4% MCDB	32.1	32.0	46.4	70.4	77.2	83.2	82.7	83.1	81.7	66.4	46.7	32.7			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	28.5	28.7	36.2	54.2	60.2	66.3	67.8	66.1	59.3	50.8	40.5	30.8			
	2% MCDB	29.2	29.4	37.8	66.2	74.9	81.5	80.7	82.2	75.0	62.1	43.3	31.4			
	5% WB	24.7	25.4	34.1	51.6	58.3	64.6	66.5	64.6	56.8	48.3	37.2	28.0			
	5% MCDB	25.3	26.2	35.4	63.0	73.1	79.5	79.9	80.1	71.9	57.5	39.1	28.6			
	10% WB	21.3	22.1	32.3	48.7	56.6	62.9	65.1	62.9	54.6	45.6	34.2	24.8			
	10% MCDB	22.2	22.9	33.7	59.4	71.1	77.4	77.8	77.5	67.1	53.8	35.9	25.8			
	MDBR	14.3	15.7	15.8	20.0	24.2	23.2	22.2	23.7	22.6	18.1	12.7	13.9			
	5% DB	14.4	13.9	12.1	29.1	30.1	28.9	28.0	30.2	30.2	28.2	14.4	14.0			
(36) Mean Daily Temperature Range	5% DB	14.4	13.9	10.3	14.0	13.1	11.0	9.9	10.9	12.8	15.3	11.4	13.9			
	5% WB	14.5	14.1	11.6	26.5	25.9	24.6	22.9	25.9	26.6	24.5	13.4	13.7			
	MCWBR	14.6	14.1	10.1	14.1	12.8	10.6	9.3	10.8	12.5	14.5	11.2	13.7			
	MCWBR	14.6	14.1	10.1	14.1	12.8	10.6	9.3	10.8	12.5	14.5	11.2	13.7			
(40) Clear-Sky Solar Irradiance	taub	0.290	0.320	0.373	0.406	0.400	0.404	0.420	0.411	0.359	0.344	0.322	0.289			
	taud	2.065	2.020	1.961	2.133	2.235	2.295	2.245	2.248	2.343	2.291	2.149	2.059			
	Ebn at Noon	215	243	252	262	270	270	263	258	260	237	201	191			
	Edh at Noon	25	35	46	44	42	40	41	39	31	27	23	21			
(44) All-Sky Solar Radiation	RadAvg	424	784	1221	1514	1877	1988	1833	1606	1184	647	300	280			
	RadStd	31	36	131	142	163	164	143	117	87	86	36	48			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page